

January 2024, CSE 106
Online on Dynamic Programming (C1/C2)
Time: 40 Minutes

A **subsequence** of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters. For example, **ace** is a subsequence of **abcde**.

A **common subsequence** of two strings is a subsequence that is common to both strings.

You are given two strings **text1** and **text2**. Design and implement an efficient algorithm with dynamic programming to determine *the length of their longest common subsequence*. If there is no **common subsequence**, return **0**.

Input / Output:

Sample Input	Sample Output
BCDAACD ACDBAB	3
AGGTAB GXTXAYB	4

Submission Guidelines: Save your `.cpp` file as `2205<3-digit-Student-ID>.cpp` and upload it on Moodle in the corresponding submission link.